METRIC

MIL-PRF-5606K
10 April 2025

SUPERSEDING
MIL-PRF-5606J
05 March 2018

PERFORMANCE SPECIFICATION

HYDRAULIC FLUID, PETROLEUM BASE; AIRCRAFT, MISSILE, AND ORDNANCE, NATO Code H-515

Inactive for new design after 29 March 1996. For new designs, use MIL-PRF-87257 or MIL-PRF-83282.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification describes the characteristics and provides the requirements for a petroleum base hydraulic fluid for use in the -54 °C to +135 °C temperature range (see 6.1). This fluid is identified by military symbol OHA and NATO Code No. H-515 (see 6.7).

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

Comments, suggestions, or questions on this document should be addressed to AFPET/PTPS, 2430 C Street, Building 70, Area B, Wright-Patterson AFB, OH 45433 or e-mailed to AFPET.PTPS@us.af.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

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2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

FEDERAL STANDARDS

FED-STD-791	Testing Method of Lubricants, Liquid Fuels, and Related Products

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-PRF-6083	Hydraulic Fluid, Petroleum Base, for Preservation and Operation
MIL-PRF-46170	Hydraulic Fluid, Rust Inhibited, Fire Resistant, Synthetic Hydrocarbon Base, NATO Code No. H-544
MIL-PRF-83282	Hydraulic Fluid, Fire Resistant, Synthetic Hydrocarbon Base, Metric, NATO Code Number H-537
MIL-PRF-87257	Hydraulic Fluid, Fire Resistant, Low Temperature, Synthetic Hydrocarbon Base, Aircraft and Missile

(Copies of these documents are available online at https://quicksearch.dla.mil.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEFENSE FORMS MANAGEMENT PROGRAM

DD Form 1718	Certification of Qualified Products

(Copies of this document are available online at https://www.esd.whs.mil/DD/.)

DEFENSE STANDARDIZATION PROGRAM OFFICE

SD-6	Provisions Governing Qualification

(Copies of this document are available online at https://quicksearch.dla.mil.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

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ASTM INTERNATIONAL

ASTM D93	Standard Test Methods for Flash Point by Pensky-Martens Closed Cup Tester
ASTM D97	Standard Test Method for Pour Point of Petroleum Products
ASTM D130	Standard Test Method for Corrosiveness to Copper from Petroleum Products by Copper Strip Test
ASTM D445	Standard Test Method for Kinematic Viscosity of Transparent and Opaque Liquids (and Calculation of Dynamic Viscosity)
ASTM D664	Standard Test Method for Acid Number of Petroleum Products by Potentiometric Titration
ASTM D892	Standard Test Method for Foaming Characteristics of Lubricating Oils
ASTM D972	Standard Test Method for Evaporation Loss of Lubricating Greases and Oils
ASTM D1298	Standard Test Method for Density, Relative Density, or API Gravity of Crude Petroleum and Liquid Petroleum Products by Hydrometer Method
ASTM D1500	Standard Test Method for ASTM Color of Petroleum Products (ASTM Color Scale)
ASTM D2603	Standard Test Method for Sonic Shear Stability of Polymer-Containing Oils
ASTM D4052	Standard Test Method for Density, Relative Density, and API Gravity of Liquids by Digital Density Meter
ASTM D4172	Standard Test Method for Wear Preventative Characteristics of Lubricating Fluid (Four-Ball Method)
ASTM D4289	Standard Test Method for Elastomer Compatibility of Lubricating Greases and Fluids
ASTM D4636	Standard Test Method for Corrosiveness and Oxidation Stability of Hydraulic Oils, Aircraft Turbine Engine Lubricants, and Other Highly Refined Oils
ASTM D5185	Standard Test Method for Multielement Determination of Used and Unused Lubricating Oils and Base Oils by Inductively Coupled Plasma Atomic Emission Spectrometry (ICP-AES)

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ASTM D5949	Standard Test Method for Pour Point of Petroleum Products (Automatic Pressure Pulsing Method)
ASTM D6304	Standard Test Method for Determination of Water in Petroleum Products, Lubricating Oils, and Additives by Coulometric Karl Fischer Titration
ASTM D6793	Standard Test Method for Determination of Isothermal Secant and Tangent Bulk Modulus
ASTM D7596	Standard Test Method for Automatic Particle Counting and Particle Shape Classification of Oils Using a Direct Imaging Integrated Tester
ASTM D7647	Standard Test Method for Automatic Particle Counting of Lubricating and Hydraulic Fluids Using Dilution Techniques to Eliminate the Contribution of Water and Interfering Soft Particles by Light Extinction
ASTM E29	Standard Practice for Using Significant Digits in Test Data to Determine Conformance with Specifications

(Copies of these documents are available from https://www.astm.org.)

INTERNATIONAL ORGANIZATION FOR STANDARDIZATION (ISO)

ISO 2230	Rubber Products – Guidelines for Storage
ISO 4406	Hydraulic fluid power — Fluids — Method for coding the level of contamination by solid particles
ISO 4407	Hydraulic fluid power – Fluid contamination – Determination of particulate contamination by the counting method using an optical microscope
ISO 11171	Hydraulic fluid power – Calibration of automatic particle counters for liquids
ISO 11218	Aerospace – Cleanliness classification for hydraulic fluids
ISO 11500	Hydraulic fluid power – Determination of the particulate contamination level of a liquid sample by automatic particle counting using the light-extinction principle

(Copies of these documents are available from https://www.iso.org.)

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SAE INTERNATIONAL

SAE AMS3217/2	Test Slabs, Acrylonitrile Butadiene (NBR-L) Low Acrylonitrile, 65 – 75
SAE AMS3400	Fluid, Reference for Testing Polyalphaolefin (PAO) Resistant Material
SAE AS598	Aerospace Microscopic Sizing and Counting of Particulate Contamination for Fluid Power Systems
SAE AS4059	Aerospace Fluid Power – Contamination Classification for Hydraulic Fluids

(Copies of these documents are available from https://www.sae.org.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Qualification. The hydraulic fluid furnished under this specification shall be products that are authorized by the qualifying activity for listing on the applicable qualified products list before contract award (see 4.2 and 6.3).

3.2 Materials. The fluid shall consist of petroleum products with additive materials to improve the low-temperature flow and viscosity-temperature characteristics, resistance to oxidation, and anti-wear properties of the finished product. A red dye shall be used for coloring.

3.2.1 Additives. There shall be no restriction on the types of materials used as additives in the fluid except for those imposed by the technical requirements of this specification. Pour point depressants may be used.

3.2.1.1 Viscosity/temperature coefficient improvers. Polymeric materials may be added to the base petroleum oil in quantities not to exceed 20 percent by weight of active ingredient in order to adjust the viscosity of the finished fluid to the values specified in 3.4.

3.2.1.2 Oxidation inhibitors. Oxidation inhibitors shall be added to the base oil in quantities not to exceed two (2) percent by weight.

3.2.1.3 Anti-wear agent. The hydraulic fluid shall not contain more than three (3) percent by weight of an anti-wear agent. Tricresol phosphate (TCP) shall not be used.

3.2.1.4 Red dye. The fluid shall contain red dye in a concentration not to exceed one (1) part of dye per 10,000 parts of oil by weight.

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3.3 Properties of petroleum base stock. The properties of the petroleum base stock used in formulating the finished fluid shall be as designated in Table I when tested as specified in 4.4.

TABLE I. Properties of petroleum base stock.

Property	Limits	ASTM Test Method
Pour point, °C (max)	-60	D97, D5949*
Flash point, °C (min)	82	D93
Acid number, mg KOH/g (max)	0.10	D664
Color, ASTM Standard (max)	No. 1	D1500
Relative density @ 15.6/15.6 °C	Report	D1298, D4052*
* Referee methods.		

3.4 Properties of finished fluid. The properties of the finished fluid shall be as specified in Table II and 3.4.1 thru 3.5 when tested as specified in 4.4.

3.4.1 Color. There shall be no readily discernible difference in the color of the finished fluid compared to the standard color when tested as specified in 4.4.1.

3.4.2 Compatibility. The hydraulic fluid shall be compatible in all concentrations with each of the fluids approved under this specification and listed on the Qualified Products List QPL No. 5606, The hydraulic fluid shall also be miscible with hydraulic fluids conforming to MIL-PRF-6083, MIL-PRF-46170, MIL-PRF-83282, and MIL-PRF-87257 fluids in all proportions from -54 °C to 135 °C. No formation of resinous gums, sludge, or insoluble material shall occur (see 4.4.2).

3.4.3 Low temperature stability. The hydraulic fluid shall show no evidence of gelling, clouding, crystallization, solidification, or separation of ingredients. The presence of a slight haze that dissipates shall not be cause for failure.

3.4.4 Relative density. The relative density shall be determined and reported at qualification but shall not be limited. The relative density of conformance samples shall fall within ± 0.008 from that obtained at qualification.

3.4.5 Shear stability. When tested as specified in 4.4.6, the percent viscosity decrease of the hydraulic fluid, measured in millimeters squared per second (mm^2/s) at 40 °C, shall not be greater than the percentage viscosity decrease of the shear stability reference fluid.

3.5 Workmanship. The workmanship shall be in accordance with (IAW) high-grade, commercial practices covering this type of material. The finished fluid shall be homogeneous and free from suspended matter, grit, or other adulteration.

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TABLE II. Properties of finished fluid.

Property	Qualification	Conformance	Limits / Requirement Paragraph	Test Method		
				FED-STD-791	ASTM (ISO)	Test Paragraph
Acid number, mg KOH/g (max)	X	X	0.20		D664	
Barium content, mg/kg (max)	X		10		D5185	
Color	X	X	3.4.1		D1500	4.4.1
Compatibility	X		3.4.2			4.4.2
Copper strip corrosion, rating (max)	X	X	2e		D130	4.4.3
Corrosiveness and oxidation stability (168 hrs @ 135 °C ± 1 °C)	X				D4636(1)	
Change in acid number, mg KOH/g (max)			0.20			
Metal specimen weight change, mg/cm^2 (max) (2)						
Aluminum			± 0.2			
Cadmium plated steel (3)			± 0.2			
Copper (4)			± 0.6			
1010 Steel			± 0.2			
Magnesium			± 0.2			
Percent change in viscosity at 40 °C			-5 to +20			
Separation of insoluble materials or gumming of the fluid			None			
Evaporation loss (6 hrs @ 71 °C), % (max)	X	X	20		D972	
Flash point, °C (min)	X	X	82		D93	

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TABLE II. Properties of finished fluid (Continued).

Property	Qualification	Conformance	Limits / Requirement Paragraph	Test Method		
				FED-STD-791	ASTM (ISO)	Test Paragraph
Foaming Characteristics @ 24 °C	X	X			D892	
Foaming tendency, mL (max) (Volume at end of five-minute blowing period)			65			
Foam stability, mL (max) (Volume at end of ten-minute settling period)			Complete collapse(5)			
Isothermal secant bulk modulus @ 40 °C and 27.6 MPa (4000 psig), MPa (psi) (min)	X		1379 (200,000)		D6793	
Low temperature stability, 72 hrs @ -54 °C ± 1 °C	X	X	3.4.3	3458		
Particulate contamination. Particle count shall be determined using one of the two automatic particle counter options below(6):		X				4.4.4
Particle count, SAE AS4059, Contamination Class(6) using automatic particle counter (max)			5	3012* (7) (8)	(11500) (7) (8)	4.4.4
Particle counting, individual channel counts & ISO Code (6) using automatic particle counter (max)			16/14/11	3012* (7) (9)	D7647 (7) (9), D7596 (7) (9), (11500) (7) (9)	4.4.4
Particle count, AS4059 Contamination Class(6) using optical microscopic analysis (max)			5		(4407) (7) (8) [AS598] * (7) (8)	4.4.4

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TABLE II. Properties of finished fluid (Continued).

Property	Qualification	Conformance	Limits / Requirement Paragraph	Test Method		
				FED-STD-791	ASTM (ISO)	Test Paragraph
Pour point, °C (max)	X	X	-60		D97, D5949*	
Relative density @ 15.6/15.6 °C	X	X	3.4.4		D1298, D4052*	
Rubber swell, standard synthetic rubber L (168 hrs @ 70 °C), %	X	X	19.0 – 30.0		D4289	4.4.5
Shear stability	X		3.4.5		D2603	4.4.6
Steel-on-steel (average wear scar), mm in diameter (max)	X	X	1.0		D4172(10)	
Viscosity, mm^2/s @	X	X			D445	
-54 °C (max)			2500			
-40 °C (max)			600			
40 °C (min)			13.2			
100 °C (min)			4.90			
Water, mg/kg (max)	X	X	100		D6304	
Workmanship	X	X	3.5			

NOTES:

*	Referee methods.
(1)	Use alternate procedure 2.
(2)	There shall be no pitting, etching, or visible corrosion on the surface of the metals viewed under magnification of 20 diameters.
(3)	A slight discoloration is permitted.
(4)	Any corrosion (discoloration) produced on the surface of the copper shall not be greater than No. 3 of the ASTM D130 copper corrosion standard.
(5)	A ring of small bubbles around the edge of the graduated cylinder shall be considered complete collapse.
(6)	Particulate contamination shall be measured as either particle count, AS4059 Contamination Class or particle counting, individual channel counts & ISO Code. Particle count via optical microscopic analysis shall only be used when a calibrated automatic particle counter is not available. Use of a digital optical microscope with measure and count software shall be the standard method used for microscopic analysis.

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TABLE II. Properties of finished fluid (Continued).

NOTES:	
(7)	The container shall be thoroughly shaken immediately prior to withdrawing the sample.
(8)	Refer to Table III for particle size ranges.
(9)	Results shall be reported for particle size, channel count, and ISO code in same format and criteria as laid out in Table IV.
(10)	Condition B.

3.6 Toxic products and formulations. The hydraulic fluid shall have no adverse effect on the health of personnel when used for its intended purpose.

4. VERIFICATION

4.1 Classification of inspections. The examination and testing of the hydraulic fluid are classified as follows:

a. Qualification inspection (see 4.2).

b. Conformance inspection (see 4.3).

4.2 Qualification inspection. Qualification inspection shall consist of testing to all the requirements specified in sections 3.3 thru 3.5, Table I, and Table II. When required by the qualifying activity, additional evaluations may be required on candidate formulations.

4.2.1 Qualification process. At the initiation of the qualification process, prospective suppliers will forward a written request for such action to the qualifying activity, Air Force Petroleum Office, AFPET/PTPS, 2430 C Street, Building 70, Area B, Wright-Patterson AFB, OH 45433 or via email AFPET.PTPS@us.af.mil. This request will contain general information on the proposed candidate material. The qualifying activity will reply in writing outlining the required information to be submitted, the governing regulations for qualified products, and the cost of testing. The information that will be submitted includes:

a. Letter stating the company will adhere to the appropriate provisions of the SD-6 on company letterhead.

b. Complete formulation of the candidate product, including chemical composition in weight percent, the manufacturer, the commercial trade name and CAS number, the purities, and use of each component.

c. Safety Data Sheet (SDS) for each component and final product. The SDS will not be older than 5 years.

d. Certified laboratory test data from the manufacturer, a commercial laboratory, or combination thereof showing quantitative results of all qualification tests required by this specification as identified in Table I and Table II.

(1) Qualification laboratory test data results covering Table II requirements for compatibility testing will only be acceptable when performed by commercial or third-party laboratory and documented on a certified COA.

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e. Verification that the System for Award Management (SAM) (CAGE Codes for the corporate office and each plant) is current. This can be viewed at https://sam.gov/.

f. Identification of the manufacturing site of the specific batch of test sample to be submitted.

g. A completed DD Form 1718.

4.2.2 Qualification inspection sample. A qualification sample shall consist of a one (1) gallon container of hydraulic fluid. The qualifying activity will request manufacturer and third-party data prior to authorizing submission of the qualification sample.

4.2.3 Retention of qualification. To retain qualification of products approved for listing on the Qualified Products List (QPL), the manufacturer will be requested to verify by certification (DD Form 1718) to the qualifying activity that its product(s) comply with the requirements of this specification. The manufacturer shall also provide a recent Certificate of Analysis (COA), to include all conformance inspection requirements (see 4.3), a current SDS, and formulation. Unless otherwise specified by the qualifying activity, the time of periodic verification by certification will be in 2 ½ year intervals from the date of original qualification and will be initiated by the manufacturer. The Government reserves the right to re-examine the qualified product whenever deemed necessary to determine that the product continues to meet any or all of the specification requirements.

4.2.4 Requalification. Requalification shall be required every five (5) years or when any change is made in the source of manufacture, purity, or composition of the fluid's base stock or additives. A minor change in the fluid formulation such as a new supplier of stock material might not need requalification testing, but only after notification to, and approval by the qualifying activity (see 6.3). The documentation required for requalification are the same as requirements stated in section 4.2.1 and shall include product formulation, safety data sheet (SDS) for each component and final product, a COA showing compliance to all of the qualification requirements specified in Table I and Table II, and a one (1) gallon requalification sample from a recent production run within the previous 60 days. The SDS will not be older than five (5) years. The qualifying activity may, at its discretion, waive complete requalification or may require only partial requalification testing to determine the significance and acceptability of the proposed formulation change.

4.3 Conformance inspection. The conformance inspection (Government Lot Acceptance Testing) shall be conducted on samples from randomly selected containers of each packaged lot (see 6.4.1) and consist of tests to verify conformance to the requirements in Table II with the following exceptions:

(1) Barium

(2) Compatibility

(2) Corrosion and oxidation stability

(3) Isothermal secant bulk modulus

(4) Shear stability

(5) Storage stability

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4.4 Methods of inspection. Testing shall be performed in accordance with the methods specified in Table I, Table II, and section 4. Physical and chemical values, specified in section 3, apply to the arithmetic average of the determinations made on the samples for those values which fall within any stated repeatability or reproducibility limits of the applicable test method. Requirements contained in Table I and Table II are absolute, as defined in ASTM E29, and shall not be subject to correction for test tolerances. If multiple determinations are made, results falling within any specified repeatability and reproducibility tolerances may be averaged. For rounding off significant figures, ASTM E29, Absolute Method, shall apply to all tests required by this specification.

4.4.1 Color of the finished product. The color of the hydraulic fluid shall be compared to a standard sample prepared by adding one (1) part dye to 10,000 parts of an oil no darker than ASTM D1500 color number 1.0.

4.4.2 Compatibility. Samples of candidate fluid in amounts of 20, 100, and 180 mL shall be mixed with samples from each of the fluids previously approved under this specification and listed in Qualified Product List QPL No. 5606. Total volume of each mixture shall be 200 mL. Mixtures shall be prepared in 250 mL stoppered flasks. The flasks shall be thoroughly agitated and then stored in an oven at 135 °C for 2 hours. At the end of this time, none of the mixtures shall show any sign of sediment, haziness, or crystallization. The samples shall then be stored at -54 °C for a period of 2 hours. A slight haze that dissipates will be permitted in the samples at this time. Compatibility tests described herein shall also be conducted with a representative fluid qualified and listed on the respective Qualified Products List for MIL-PRF-6083, MIL-PRF-46170, MIL-PRF-83282, and MIL-PRF-87257.

4.4.3 Copper strip corrosion. The following procedures shall be followed:

a. Prepare one copper strip in accordance with ASTM D130.

b. Fill a test tube or other suitable container with 90 mL of hydraulic fluid and individually immerse the copper strip into the test tube.

c. Immerse the test tube, which will be equipped with an air condenser, in a constant temperature bath capable of maintaining the fluid temperature at 135 °C ± 1 °C. If an oven is used, vent the air condenser to the outside of the oven. After 72 hours at this test temperature, remove the copper strip from the fluid, rinse it in isooctane, and compare the results with the ASTM copper strip corrosion standards.

4.4.4 Particulate contamination. Particle count shall be determined as follows:

a. An automatic particle counter employing either the light extinction principle or optical imaging and meeting the channel, particle size range, and calibration requirements specified in Table II or Table III shall be used (use the operating instructions for the applicable instrument). The instrument shall meet the calibration requirements of ISO 11171.

    (1) When using an automatic particle counter, the instrument used shall have its particle cleanliness level verified prior to sample analysis. For differential particle count, the APC must meet SAE AS4059 Contamination Level 00 (Table 1) limits. For cumulative particle count, the particle counter shall meet SAE AS4059 Contamination Level 00 (Table 2) limits.

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b. Particle counting via microscopic analysis shall only be used if a calibrated automatic particle counter is not available. Use of a digital optical microscope with measure and count software shall be the standard method used for microscopic analysis.

c. Mix the sample thoroughly immediately prior to withdrawing the sample.

TABLE III. Particle Count, SAE AS4059/ISO 11218 Contamination Level 5.

Particle Size Ranges, micrometers (µm)[1] AIA/NAS 1638[2] (ISO 11171)		Maximum number of particles per 100 mL
SAE AS4059	SAE AS4059	DIFFERENTIAL
5-15	6-14(c)	8000
15-25	14-21(c)	1425
25-50	21-38(c)	253
50-100	38-70(c)	45
>100	>70(c)	8
SAE AS4059	SAE AS4059	CUMULATIVE
>5	>6(c)	9731
>15	>14(c)	1731
>25	>21(c)	306
>50	>38(c)	53
>100	>70(c)	8

NOTES:

1. The symbol (c) is used to designate the automatic particle counter was calibrated to ISO 11171.
2. AIA/NAS 1638 - Cleanliness Requirements of Parts Used in Hydraulic Systems is obsolete and replaced with SAE AS4059. Referenced for transition purposes only.

TABLE IV. Particle Count, ISO 4406 (SAE AS4059/ISO 11218 Contamination Level 5 Basis).

Particle Size Ranges, micrometers (µm) [1]		
Size	Counts per mL	ISO Code [3]
≥ 4 µm(c)	[2]	Max 16
≥ 6 µm(c)	[2]	Max 14
≥ 14 µm(c)	[2]	Max 11
≥ 21 µm(c)	[2]	[2]
≥ 38 µm(c)	[2]	[2]
≥ 50 µm(c)	[2]	[2]
≥ 70 µm(c)	[2]	[2]

NOTES:

1. The symbol (c) is used to designate the automatic particle counter was calibrated to ISO 11171.
2. Report.
3. Per Table 1 “Allocation of scale numbers” of ISO 4406.

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4.4.5 Rubber swell. Rubber swell shall be performed IAW ASTM D4289 with the following modifications:

a. Use the standard synthetic rubber, NBR-L meeting the latest revision of SAE AMS3217/2. Store in the dark, in an inert atmosphere, below 65% relative humidity, away from direct heat sources, and between 15 °C to 25 °C (59 °F to 77 °F) IAW ISO 2230.

b. Check elastomer for conformance to SAE AMS3217/2, latest revision, at least every 6 months from the date of manufacture using SAE AMS3400 as the reference fluid.

c. Age the specimens by storing the container in a convection oven at 70 °C ± 2.5 °C for 168 hrs ± 0.5 hrs.

4.4.6 Shear stability. Shear stability tests shall be conducted IAW ASTM D2603, with the following modifications:

a. Control fluid is ASTM Reference Fluid B.

b. Run the control fluid and the test fluid consecutively in the same apparatus and under the same test conditions for 30 minutes at 0 °C. The equipment is to be such that the viscosity decrease of the reference fluid, at 40 °C is approximately 15 percent. Use 30 mL of fluid for this test.

4.5 Examination of filled containers. A random sample of filled containers from each packaged lot (see 6.4.1), shall be examined with regard to fill closure, sealing and leakage.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The hydraulic fluid covered by this specification is intended for use in automatic pilots, shock absorbers, brakes, flap-control mechanisms, missile hydraulic servo-controlled systems, and other hydraulic systems which use synthetic sealing material.

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6.1.1 Storage conditions. Prior to use in the intended equipment, the product may be stored under conditions of covered or uncovered storage in geographic areas ranging in temperatures from -57 °C to +49 °C (-70 °F to +120 °F).

6.1.2 Interchangeability. This fluid is completely compatible with MIL-PRF-6083, MIL-PRF-46170, MIL-PRF-83282, and MIL-PRF-87257 hydraulic fluids. It may be interchangeable for some applications. The selection of the fluids to be used depends on the requirements of the operational system.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of the specification.

b. Type, and capacity of containers.

c. Quantity.

d. Packaging requirements (see 5.1).

6.3 Qualification. With respect to products that require qualification, awards will be made only for products that are, at the time of award of contract, qualified for inclusion in Qualified Products List QPL No. 5606 whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or purchase orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from the Air Force Petroleum Office, AFPET/PTPS, 2430 C Street, Bldg. 70, Area B, Wright-Patterson AFB, OH 45433; e-mail: AFPET.PTPS@us.af.mil. An online listing of products to this specification may be found in the Qualified Products Database (QPD) at https://assist.dla.mil.

6.3.1 Qualification information. It is understood that the material furnished under this specification subsequent to final approval will be of the same composition and will be equal to products on which approval was originally granted. In the event that fluid furnished under contract is found to deviate from the composition of the approval product, or that the product fails to perform satisfactorily, approval of such products may be subjected to immediate withdrawal from the QPL at the discretion of the qualifying activity.

6.3.2 Proceed authorization notification. Upon successful review of the submitted information, the Air Force Petroleum Office will issue authorization to ship the qualification sample(s) for testing.

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6.3.3 Qualification samples. Upon receiving authorization from the Air Force Petroleum Office, qualification sample(s) may be shipped to the qualifying activity, as detailed in the authorization letter. Sample(s) should be accompanied by an SDS and a copy of the authorization letter. Each sample should be plainly identified by securely attached durable tags or labels marked with the following information:

a. QUALIFICATION SAMPLE
b. Hydraulic Fluid, Petroleum Base, Aircraft, Missile, and Ordnance
c. MIL-PRF-5606
d. Name of Manufacturer
e. Product Code Number
f. Batch Number
g. Date of Manufacture

6.4 Definitions.

6.4.1 Packaged lot. A packaged lot is defined as an indefinite number of unit containers of identical size and type offered for acceptance, and filled with a homogeneous mixture of material manufactured by a single plant run (not exceeding 24 hours) through the same processing equipment, with no change in ingredient material

6.5 Shelf-life. This specification covers items where the assignment of a Federal shelf-life code is a consideration. Specific shelf-life requirements should be specified in the contract or purchase order, and should include, as a minimum, shelf-life code, shelf-life package markings IAW MIL-STD-129 or FED-STD-123, preparation of a materiel quality storage standard for type II (extendible) shelf-life items, and a minimum of 85 percent shelf-life remaining at time of receipt by the Government. These and other requirements, if necessary, are in DoDM 4140.27, Volume I, *Shelf-Life Management Program: Program Administration;* and Volume 2, *Shelf-Life Management Program: Materiel Quality Control Storage Standards.* The shelf-life codes are in the Federal Logistics Information System Total Item Record. Additive information for shelf-life management may be obtained from DoDM 4140.27 Volumes 1 and 2, or the designated shelf-life Points of Contact (POC). The POC should be contacted in the following order: (1) the Inventory Control Points that manage the item and (2) the DoD Service and Agency administrators for the DoD Shelf-Life Program. Appropriate POCs for the DoD Shelf-Life Program can be contacted through the DoD Shelf-Life Management website: https://www.shelflife.dla.mil/.

6.6 Subject term (key word) listing.

Additive
Anti-wear
Corrosion
Flash point
Isothermal secant bulk modulus
Lubricant
Oxidation
Pour point
Shear stability
Tricresyl phosphate
Viscosity

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6.7 International standardization agreement implementation. This specification implements:

a. Air Force Interoperability Council (AFIC) Air Standard Fuels, Lubricants and Gases (FLG) 4023, *Hydraulic Fluid, Petroleum (H-515) and Polyalphaolefin (H-537 and H-538)*.

b. AFIC Air Standard FLG 4024, *Interchangeability Chart of Standardised Aviation Fuels, Lubricants, and Associated Products*.

c. NATO STANAG/AFLP-1135, *Interchangeability of Fuels, Lubricants and Associated Products Used by the Armed Forces of the North Treaty Nations*.

d. NATO STANAG/AFLP-3748, *Hydraulic Fluid, Petroleum (H-515), Polyalphaolefin (H-537 and H-538) and Phosphate Ester Fluids (H-522, H-523, H-524)*.

When amendment, revision, or cancellation of this specification is proposed, the preparing activity will must coordinate the action with the U.S. National Point of Contact for the international standardization agreement, as identified in the ASSIST database at https://assist.dla.mil.

6.8 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

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CONCLUDING MATERIAL

Custodians:
Army – AT
Navy – AS
Air Force – 68

Preparing activity:
Air Force – 68
(Project 9150-2025-001)

Review activities:
Army – AR, MI, SM
Navy – OS, SA, SH
Air Force – 20
DLA – GS
DTRA – DS

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information using the ASSIST Online database at https://assist.dla.mil.

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